

Mustang Math Tournament 2025

Herding Hexes Stallion Round



Basic Format

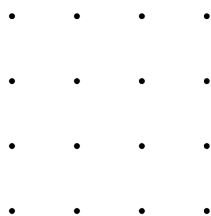
- This round contains 26 problems to be solved in 45 minutes.
- Each problem corresponds to a hexagon on the answer grid (backside).
- A correct answer will grant 2 points each for Problems 1 through 10, 3 points each for Problems 11 through 19, and 4 points each for Problems 20 through 26.
- The score of a hexagonal tile is doubled if it can be connected back to the Free tile through other tiles that contain correct answers.
- **Do not** write below the provided answer blank inside each hexagon (the space is for grading purposes).
- *Feel free to flag down a proctor if you need help deciphering any of the above instructions*

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**WE WILL NOT GRADE ANY WORK ON THIS
PAGE.**

1. A rectangular TV has width w , length l , and diagonal h . TV measurements are given in terms of the diagonal length, not the length. If a TV is advertised to be $h = 25$ inches and $w = 20$ inches, what is l (in inches)?
2. Let $a \oplus b$ equal $\sqrt{a^2 + b^2}$. Calculate $(12 \oplus 3) \oplus 4$.
3. What is the number of divisors of 2025 which are multiples of 5?
4. You are at a lottery with a prize of \$27,000 and only one winning ticket. There are 5,400 tickets and you buy 70 of them. What are your expected winnings?
5. Compute the remainder when 5^{390539} is divided by 2023.
6. Max has 5 blocks with the numbers 1, 2, 3, 4, and 5 arranged in increasing order from left to right. Then, he closes his eyes and mixes all the blocks up randomly and rearranges them in a row in a random order. Given that the number on the leftmost block is prime, find the probability that the number on the second leftmost block is also prime. Express your answer as a common fraction.
7. Define $f(x) = x^2 + 40$, and let m be a positive real number such that $f(2m) = 2f(m)$. Find m^2 .
8. Equilateral triangle ABC with side length 3 is inscribed inside circle O . Another circle is internally tangent to circle O at point T and sides AB and AC at points M and M' respectively. What is the area of triangle $MM'T$?
9. Let $x = (8) \times (9999\dots 9)$ where the second part has n 9's. What value of n has the sum of the digits of x equal to 999?
10. What is the smallest positive integer n such that 24 divides $n^2 + 7n$?
11. Tristan flips a coin 22 times. For the first 2 flips, he gets 1 heads and 1 tails. For every flip after the second, the Evil Yuuki swaps out Tristan's coin with an unfair coin such that after n flips, the probability that the next flip is heads is $\frac{h}{n}$ where h is the number of heads Tristan had gotten up until that point. Compute the probability that Tristan's last 20 flips all landed tails. Express your answer as a common fraction.
12. Arpit is running in a straight line at 60 miles per hour. He has a watch, and the time is 1:00 AM. Every 10 minutes, he checks his watch, making sure that the 12 points in the direction he is currently running, then changes his direction to run in the direction of the minute hand. Also, since Arpit gets tired, he slows down by 10 miles per hour every time he checks his watch. When Arpit stops, how far away from his starting point will he be?
13. Let a and b be integers that correspond to $(6a + b)^2 - (a + 6b)^2 = 4900$. If $ab \geq 0$, what is the smallest value of ab ?

14. In the image below, there are 16 points, forming a 4 by 4 lattice. How many triangles exist with vertices each being one of the 16 points?



15. Benjamin is sick with polynomialitis after doing a very joyful problem. His doctor prescribes him medicine in the shape of a capsule of width 0.25 inches and height 1 inch. Given that a capsule consists of two perfect hemispheres on the ends of a cylinder, the volume of the pill can be expressed as $\frac{a}{b}\pi \text{ in}^3$ for positive relatively prime integers a and b . What is $a + b$?
16. What is the smallest possible integer numerator for a fraction with a denominator of 34650 that can be expressed as a positive terminating decimal?
17. Find the units digit of $|2^8 + 3^8 - 5^9|$.
18. A bag has 121 balls, either black or white. The probability of drawing 2 balls of the same color in a row, without replacement, is $\frac{1}{2}$. Find the absolute value of the amount of black balls minus the amount of white balls.
19. Let a and b be digits in base $(a + b)$ such that the value of the 2-digit numeral $\underline{ab}$ in base $(a + b)$ is equal to 144 in base 10. What is $a + b$ in base 10?
20. The area of a right triangle is 6 times the sum of the two leg lengths. How many non-congruent right triangles with integer leg lengths satisfy these properties?
21. Alice is on the 20th floor of a 101-story building (including the ground floor). Every minute, Alice flips a fair coin. If the coin lands on heads, Alice moves up a level. If the coin lands on tails, Alice moves down a level. Once Alice reaches either the ground floor or the 100th floor, she will stop. What is the probability that Alice stops on the 100th floor? Express your answer as a simplified common fraction.
22. Bob is hanging out with his martian friends, all of whom use different number systems. Bob writes down a 3-digit number. The first friend, who interprets it in base 5, thought it meant 48 base 10. The second friend, who works in base 6, thought it was equal to 63 base 10. The final friend in base 7 thought it was 80 base 10. What is the sequence of digits?
23. Tristan receives 4 test scores, all integers between 1 and 10,000, inclusive. After he arranges them in nondecreasing order, he notices that the list of numbers is a geometric series with common ratio 2. Furthermore, the average of the first i numbers is an integer, for all i from 1 to 4, inclusive. How many possibilities are there for Tristan's scores, in the order that he received them from the teacher?

24. Let $f(x) = \frac{(x+1)^2}{x+2}$ and $g(x) = \frac{(x+1)^2(x+3)^2}{(x+2)^2}$.
If a and b are the solutions of the equation $f(x) = 4$, what is the value of $(7g(a) + 40)(7g(b) + 40)$?
25. Let $ABCD$ be a rectangle with $AB = 9$ and $BC = 4$. Points P, Q, R , and S lie on sides AB, BC, CD , and DA , respectively. Points E, F, G , and H lie on SP, PQ, QR , and RS , respectively, such that $EFGH$ is a rectangle with $EF = 5$, $FG = 2$, and $EF \parallel AB$. What is the area of $PQRS$?
26. Let S be the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. Find the number of sequences $A_1, A_2 \cdots A_{16}$ of distinct subsets of S such that for all positive integers k , A_k and A_{k+1} share no elements, and if k is odd, then A_k and A_{k+1} have a combined size of 8.

Acceptable Answers

The following rules provide guidelines for acceptable answers in this round. Please note that any specifications provided in a problem will take precedence over these rules. The decisions of MMT coordinators are final.

- Common fractions are defined as a fraction in the form $\pm \frac{a}{b}$ where a and b are natural numbers and $\gcd(a, b) = 1$.
- Ratios and fractional answers should be expressed as common fractions unless otherwise specified.
- Radicals should be simplified. A simplified radical must satisfy:
 - No square factors, fractions, or nested radicals inside a radical
 - No radicals inside the denominator of a fraction
- Answers must be expressed to the exact accuracy called for in the problem (e.g. 25.0 will not be accepted for 25 and 25 will not be accepted for 25.0).
- Do not make approximations for numbers (e.g. 3.14 or $\frac{22}{7}$ for π) unless otherwise specified.
- Units **do not** need to be included but must be correct if included.